Nikhil Reddy

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Education

Rutgers University Honors College – New Brunswick, NJ Expected May 2029
Bachelors in Electrical Engineering
- GPA: 3.9 / 4.0

Projects

FPGA-Accelerated Handheld Game System | *Personal Project* Ongoing
- Programmed a Lattice ECP5 FPGA to act as a hybrid 2D/3D GPU (custom scanline PPU, 3D rasterizer, compositor) paired with an STM32H723 CPU over an FMC/AXI-Stream bus
- Pipelined the entire rendering & scanline cycle of the GPU to clock 66+ MHz on a DS game rewritten from ARM9 to fit on ARM Cortex-M hardware
- Designed an SDRAM-AXI memory controller to sustain native, non-emulated rendering at a stable 60 FPS

All-Band, All-Mode Transceiver | *Personal Project* Ongoing
- Utilized FCC Technician Radio License (Callsign: KE2IVS) to operate voice, data, and CW communications across HF/VHF/UHF amateur radio bands
- Designed a universally accommodating transceiver architecture with swappable band-pass filter modules, allowing for wide frequency range
- Built a fully bias/drive-tunable power amplifier system capable of transmission up to 6W on SSB/FM/CW/FT8

Experience

Undergraduate Research Assistant, WINLAB, Rutgers University Sept 2026 – Present
- Configured high-performance Tabor Proteus PCIe Arbitrary Waveform Transceiver for high-throughput, wideband SDR testing environments
- Developed HIL automation scripts to simulate digital IQ waveforms and their degradation in non-ideal atmospheric conditions at sampling rates up to 9 Gs/s

Electrical Lead, Rutgers IEEE Intelligent Ground Vehicle Competition Sept 2025 – Present
- Built an autonomous waterproofed rover with cooling, suspension, and onboard computer vision to navigate a 2000 ft course with obstacles and uneven terrain
- Directed development of power systems, CAN communication lines, peripherals, safety shutdown protocols, and other electrical subsystems within the rover

Avionics Subteam, Hardware Engineer, Rutgers AIAA Rocket Propulsion Laboratory Sept 2025 – Present
- Constructed a multi-stage rocket with radio, cargo, and onboard avionics systems for the FAR and IREC Competitions, coordinating electrical integration across recovery, telemetry, and payload subsystems
- Upgraded flight computers for liquid rocket engines, developed an altimeter for real-time 0.2W transmission during launch, and designed a Bluetooth remote arming system per new range regulations

Electrical Team Engineer, Rutgers University Dynamics June 2026 – Present
- Designed power distribution and battery management hardware for a 4×6S LiPo pack, integrating a BQ76952 battery monitor with an STM32-based MCU for cell balancing, current sensing, and CAN-based telemetry
- Developed embedded firmware for thermal management and system-level power control, including reverse-polarity protection, precharge sequencing, and an active regen clamp

Skills

Languages: Java, Python, C, C++, Verilog, MATLAB/Simulink
Software & Simulation: TensorFlow, Scikit-Learn, Verilator, ROS
Electrical: Analog Circuits, PCB Design, RF, Embedded Systems, STM32, CAN, SPI, I2C
Design & Dev Tools: Altium, EasyEDA, KiCad, Lattice Diamond, VS Code, Yosys, Proteus AWG
Certifications: FCC Technician Radio License (KE2IVS), JLCPCB Ambassador